

Let, x be an arbitrary element of A , i.e. $x \in A$.

By Defn. of OR, $x \in A \vee x \in B$. By Defn. 1, $x \in A \cup B$. By Defn. 2, $x \in A \wedge x \in A \cup B$. By Defn. of AND, $x \in A \cap (A \cup B)$.

By Defn. 3 of Section 2.1, $A \subseteq A \cap (A \cup B)$

$$A \cap (A \cup B) = A \quad \square$$

14) Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$ and $A \cap B = \{3, 6, 9\}$.

Solution:

By Defn. 4, $A - B$ represents: $\{x | x \in A \wedge x \notin B\}$.

By Defn. 2, $A \cap B = \{x | x \in A \wedge x \in B\}$.

$$\begin{aligned} (A - B) \cup (A \cap B) &= (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in B) \quad (\text{By Defn. 1}) \\ &= (x \in A \wedge (x \notin B \vee x \in B)) \quad (\text{2nd Distributive Law of Propositions}) \end{aligned}$$

$$= (x \in A \wedge (x \in \bar{B} \vee x \in B)) \quad (\text{By Defn. 5})$$

$$= (x \in A \wedge (x \in B \vee x \in \bar{B})) \quad (\text{By 1st Commutative Law of Propositions})$$

$$= (x \in A \wedge (x \in B \cup \bar{B})) \quad (\text{Defn. 1})$$

$$= (x \in A \wedge x \in U) \quad (\text{Prob 9-(a)})$$

$$= x \in A \cap U \quad (\text{Defn. 2}) = x \in A \quad (\text{Prob 6(b)})$$

$$\therefore A = \{1, 3, 5, 6, 7, 8, 9\}$$

$$\text{Similarly, } B = (B - A) \cup (A \cap B) = \{2, 3, 6, 9, 10\} \quad \square$$

15) Prove the second De Morgan law in Table 1 by showing that if A and B are sets, then $\overline{A \cup B} = \bar{A} \cap \bar{B}$

a) by showing each side is a subset of the other side.

b) using a membership table.

Solution: a) Let, x be an arbitrary element of $\overline{A \cup B}$, i.e. $x \in \overline{A \cup B}$.

By Defn. 5, $x \notin A \cup B$. By Defn. of not belongs to, $\neg(x \in A \cup B)$ is True. By Defn. 1, $\neg(x \in A \vee x \in B)$ is True. By 2nd De Morgan's Law of Propositions, $\neg(x \in A) \wedge \neg(x \in B)$ is True. By Defn. of not belongs to, $(x \notin A \wedge x \notin B)$ is True. By Defn. 5, $(x \in \bar{A} \wedge x \in \bar{B})$ is True. By Defn. 2, $(x \in \bar{A} \cap \bar{B})$. By Defn. 3 of Sec-2.1, $\overline{A \cup B} \subseteq \bar{A} \cap \bar{B}$